In [1]:
```
import yaml
import matplotlib.pyplot as plt
from scipy.stats import skew, kurtosis, ks_2samp
from scipy.stats import wasserstein_distance as w1dist, pearsonr
from factor_model import load_model
from scenario_generator import *

with open("cfg.yaml") as f:
    _exp = yaml.safe_load(f)["experiment_name"]
PREFIX = f"model/{_exp}"
with open(f"{PREFIX}/cfg.yaml") as f:
    _cfg = yaml.safe_load(f)

NUM_GENERATE = _cfg["num_generate"]
rng = np.random.default_rng(42)
```

In [2]:
```
train_model = load_model(f"{PREFIX}")
test_model = load_model(f"{PREFIX}/test")
print(PREFIX)
```

```
model/DDPM
```

In [3]:
```
R_df = pd.read_parquet(train_model.data_source)
R_testdf = pd.read_parquet(test_model.data_source)
common_stocks = np.intersect1d(R_df['csecid'].unique(), R_testdf['csecid'].u
col_indices_test = np.searchsorted(np.sort(R_testdf['csecid'].unique()), com
col_indices = np.searchsorted(np.sort(R_df['csecid'].unique()), common_stock
R_testdf = R_testdf[R_testdf['csecid'].isin(common_stocks)]
R_df = R_df[R_df['csecid'].isin(common_stocks)]
R_train = R_df.pivot_table(index="date", columns="csecid", values="returns")
R_test = R_testdf.pivot_table(index="date", columns="csecid", values="returr
```

In [4]:
```
"""
R_df = pd.read_parquet(train_model.data_source)
assert np.array_equal(R_train.columns[col_indices], common_stocks), "Column
R_train = R_train.values
# test if the common stocks are correctly aligned
R_train = R_df.pivot_table(index="date", columns="csecid", values="returns")
R_test = R_testdf.pivot_table(index="date", columns="csecid", values="returr
assert R_test.columns.equals(R_train.columns), "Columns do not match common
"""
```

Out[4]:
```
'\nR_df = pd.read_parquet(train_model.data_source)\nassert np.array_equal(R
_train.columns[col_indices], common_stocks), "Column indices do not match c
ommon stocks"\nR_train = R_train.values\n# test if the common stocks are co
rrectly aligned\nR_train = R_df.pivot_table(index="date", columns="csecid",
values="returns")\nR_test = R_testdf.pivot_table(index="date", columns="cse
cid", values="returns")\nassert R_test.columns.equals(R_train.columns), "Co
lumns do not match common stocks"\n'
```

In [5]:
```
diffusionScenario = ScenarioGenerator(train_model, DiffusionSampler(f"{PREFI
R_gen = diffusionScenario.stock_generate(NUM_GENERATE)[:, col_indices]
R_gen.shape
```

```
/home/renyi/anaconda3/envs/diffusion_factor_model/lib/python3.12/site-packag
es/tqdm/auto.py:21: TqdmWarning: IProgress not found. Please update jupyter
and ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/user_instal
l.html
  from .autonotebook import tqdm as notebook_tqdm
```

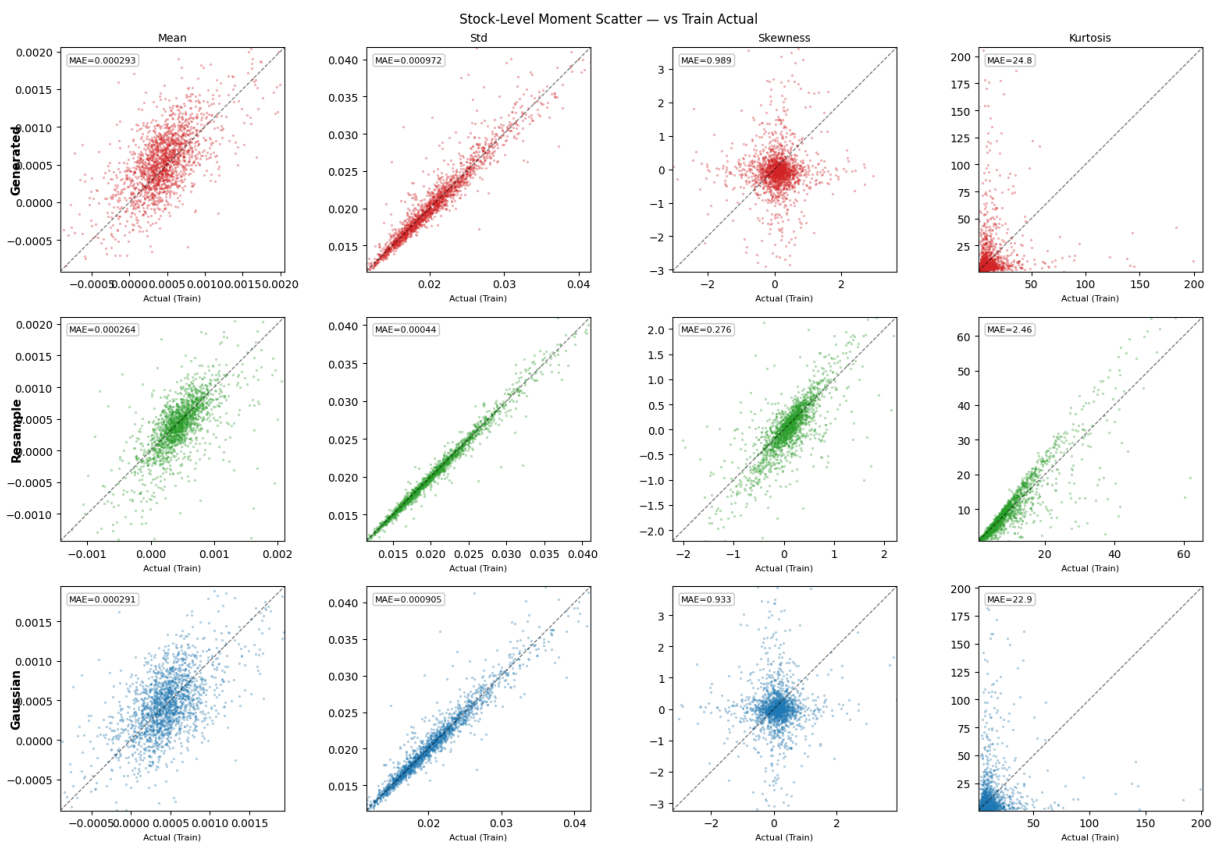
Out[5]: (4096, 1924)

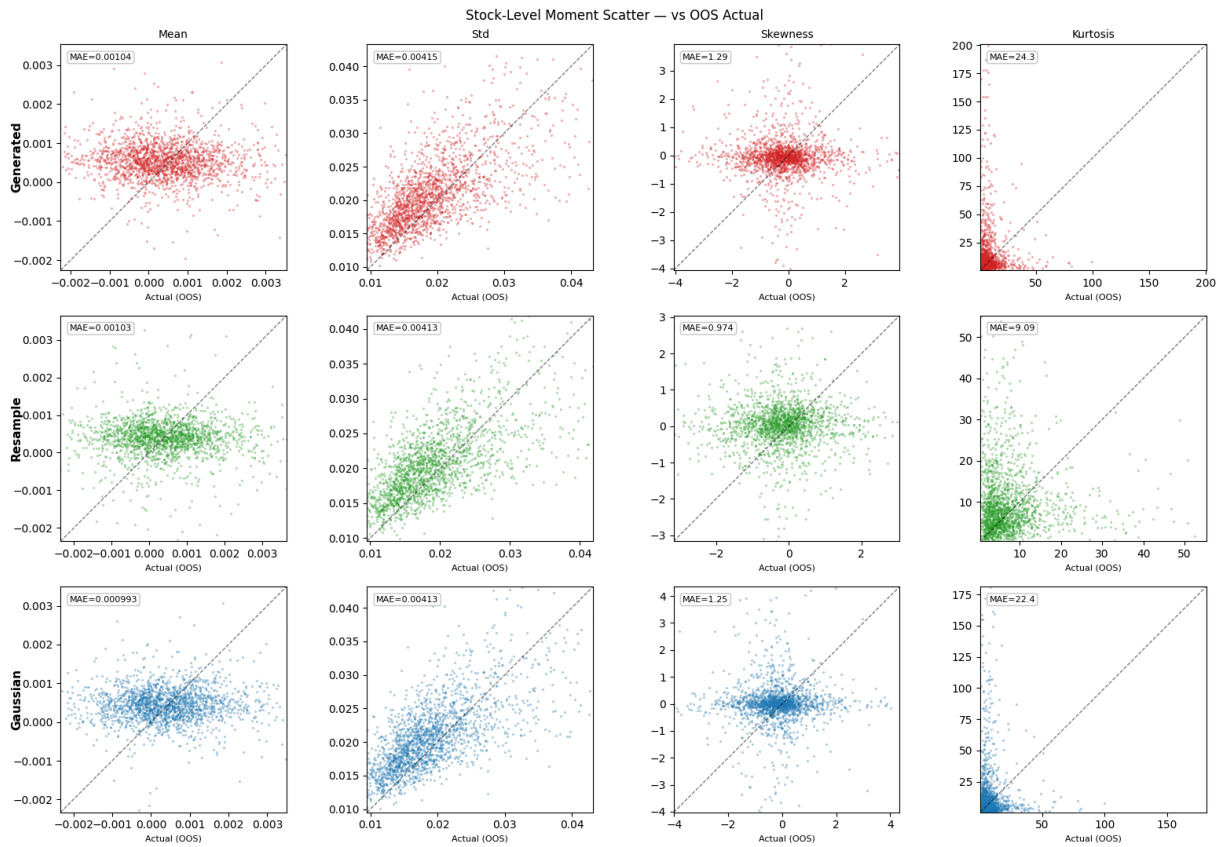
In [6]:
```
# get the baseline sample after filtering
row_idx = rng.choice(R_train.shape[0], size=NUM_GENERATE, replace=False)
R_rs = R_train[row_idx, :]
R_test = R_testdf.pivot_table(index="date", columns="csecid", values="returr
gaussionScenario = ScenarioGenerator(train_model, GaussianSampler(train_mode
R_gaussian = gaussionScenario.stock_generate(NUM_GENERATE)[:, col_indices]

print(f"Train returns: {R_train.shape}  NaN fraction: {np.isnan(R_train).mea
print(f"Test returns:  {R_test.shape}   NaN fraction: {np.isnan(R_test).mear
print(f"Resample returns: {R_rs.shape}  NaN fraction: {np.isnan(R_rs).mean()
print(f"Gaussion returns: {R_gaussian.shape} NaN fraction: {np.isnan(R_gauss
```

```
Train returns: (6262, 1924)  NaN fraction: 39.33%
Test returns:  (261, 1924)   NaN fraction: 4.27%
Resample returns: (4096, 1924)  NaN fraction: 39.35%
Gaussion returns: (4096, 1924) NaN fraction: 0.00%
```

In [7]:
```
## Stock-Level Moment Scatter Plots
MOMENTS = {
    "Mean":     lambda X: np.nanmean(X, axis=0),
    "Std":      lambda X: np.nanstd(X, axis=0),
    "Skewness": lambda X: skew(X, axis=0, nan_policy="omit"),
    "Kurtosis": lambda X: kurtosis(X, axis=0, nan_policy="omit"),
}

m_actual_tr  = {k: fn(R_train) for k, fn in MOMENTS.items()}
m_actual_oos = {k: fn(R_test)  for k, fn in MOMENTS.items()}
m_gen        = {k: fn(R_gen)       for k, fn in MOMENTS.items()}
m_rs         = {k: fn(R_rs)        for k, fn in MOMENTS.items()}
m_gs         = {k: fn(R_gaussian)  for k, fn in MOMENTS.items()}

MODELS = [
    ("Generated", m_gen, "#d62728"),
    ("Resample",  m_rs,  "#2ca02c"),
    ("Gaussian",  m_gs,  "#1f78b4"),
]

def _moment_scatter(actual_label, m_actual, fig_title):
    fig, axes = plt.subplots(3, 4, figsize=(16, 11))
    for row, (model_label, m_pred, color) in enumerate(MODELS):
        for col, (mname, _) in enumerate(MOMENTS.items()):
            ax = axes[row, col]
            x = m_actual[mname]
            y = m_pred[mname]
            mask = np.isfinite(x) & np.isfinite(y)

            lo = np.percentile(np.concatenate([x[mask], y[mask]]), 1)
            hi = np.percentile(np.concatenate([x[mask], y[mask]]), 99)
```

```
            ax.scatter(x[mask], y[mask], s=5, alpha=0.35, linewidths=0, colo
            ax.plot([lo, hi], [lo, hi], "k--", lw=0.9, alpha=0.55)

            mae = np.mean(np.abs(x[mask] - y[mask]))
            ax.text(0.04, 0.96, f"MAE={mae:.3g}",
                    transform=ax.transAxes, fontsize=8, va="top", ha="left",
                    bbox=dict(boxstyle="round,pad=0.25", facecolor="white",
                              edgecolor="0.75", alpha=0.85))

            ax.set_xlim(lo, hi)
            ax.set_ylim(lo, hi)
            ax.set_aspect("equal")
            ax.set_xlabel(f"Actual ({actual_label})", fontsize=8)
            if row == 0:
                ax.set_title(mname, fontsize=10)

        axes[row, 0].annotate(model_label, xy=(-0.22, 0.5), xycoords="axes f
                              fontsize=11, rotation=90, va="center", fontwei
    fig.suptitle(fig_title, fontsize=12)
    fig.tight_layout()
    plt.show()

_moment_scatter("Train", m_actual_tr,  "Stock-Level Moment Scatter — vs Trai
_moment_scatter("OOS",   m_actual_oos, "Stock-Level Moment Scatter — vs OOS
```





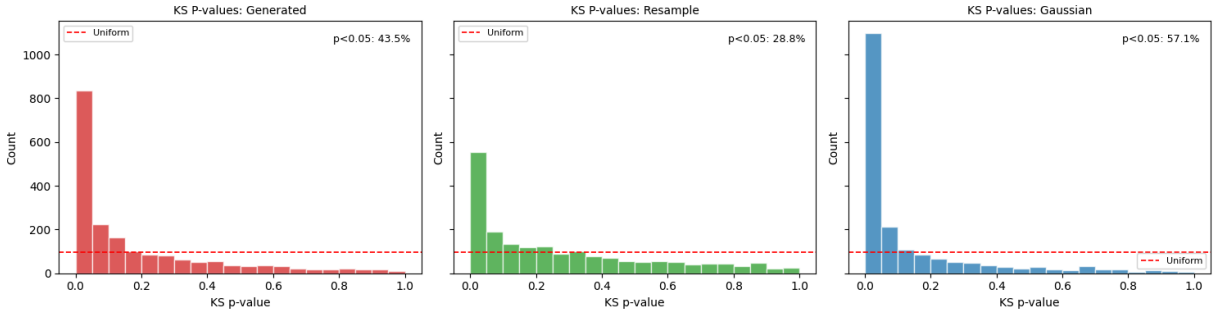
In [8]:

```
col_w = 14
header = f"{'':22}" + "".join(f"{m:<{col_w}}" for m in MOMENTS)
print("MAE (cross-stock mean absolute error)")
print(header)
print("-" * len(header))
for actual_label, m_actual in [("Train", m_actual_tr), ("OOS", m_actual_oos)
    for model_label, m_pred in [("Generated", m_gen), ("Resample", m_rs), ("
        label = f"{actual_label} {model_label}"
        row = f"{label:<22}"
        for mname in MOMENTS:
            x, y = m_actual[mname], m_pred[mname]
            mask = np.isfinite(x) & np.isfinite(y)
            mae = np.mean(np.abs(x[mask] - y[mask]))
            row += f"{mae:<{col_w}.4g}"
        print(row)
    print()
```

```
MAE (cross-stock mean absolute error)
                       Mean          Std           Skewness      Kurtosis
-----------------------------------------------------------------------------
--
Train Generated        0.000293      0.0009715     0.9892        24.81
Train Resample         0.0002637     0.0004404     0.2756        2.464
Train Gaussian         0.0002913     0.0009051     0.933         22.86

OOS Generated          0.001037      0.004147      1.295         24.34
OOS Resample           0.001027      0.004129      0.9743        9.091
OOS Gaussian           0.000993      0.004131      1.253         22.4
```

In [19]:

```
## 3.2 Distribution of KS Test P-values (all models vs OOS)
pvals_gen_oos, pvals_rs_oos, pvals_gs_oos = [], [], []
S = R_train.shape[1]
for s in range(S):
    g   = R_gen[:, s];      g   = g[np.isfinite(g)]
    rs  = R_rs[:, s];       rs  = rs[np.isfinite(rs)]
    gs  = R_gaussian[:, s]; gs  = gs[np.isfinite(gs)]
    oos = R_test[:, s];     oos = oos[np.isfinite(oos)]
    if len(g)  > 5 and len(oos) > 5: pvals_gen_oos.append(ks_2samp(g,  oos).
    if len(rs) > 5 and len(oos) > 5: pvals_rs_oos.append( ks_2samp(rs, oos).
    if len(gs) > 5 and len(oos) > 5: pvals_gs_oos.append( ks_2samp(gs, oos).

pvals_gen_oos = np.array(pvals_gen_oos)
pvals_rs_oos  = np.array(pvals_rs_oos)
pvals_gs_oos  = np.array(pvals_gs_oos)

fig, axes = plt.subplots(1, 3, figsize=(15, 4), sharey=True)

n_bins = 20
y_max = max(
    np.histogram(pvals_gen_oos, bins=n_bins, range=(0, 1))[0].max(),
    np.histogram(pvals_rs_oos,  bins=n_bins, range=(0, 1))[0].max(),
    np.histogram(pvals_gs_oos,  bins=n_bins, range=(0, 1))[0].max(),
)

for ax, pv, label, color in [
    (axes[0], pvals_gen_oos, "Generated", "#d62728"),
    (axes[1], pvals_rs_oos,  "Resample",  "#2ca02c"),
    (axes[2], pvals_gs_oos,  "Gaussian",  "#1f78b4"),
]:
    ax.hist(pv, bins=n_bins, range=(0, 1), color=color, alpha=0.75, edgecolo
    ax.axhline(len(pv) / n_bins, color="red", linestyle="--", lw=1.2, label=
    ax.set_xlabel("KS p-value", fontsize=10)
    ax.set_ylabel("Count", fontsize=10)
    ax.set_title(f"KS P-values: {label}", fontsize=10)
    ax.set_ylim(0, y_max * 1.05)
    ax.legend(fontsize=8)
    frac = (pv < 0.05).mean()
    ax.text(0.97, 0.95, f"p<0.05: {frac:.1%}", transform=ax.transAxes,
            ha="right", va="top", fontsize=9)

fig.tight_layout()
plt.show()
```



In [18]:
```
col_w = 20
header = (f"{'':16}" + f"{'Generated':<{col_w}}"
          + f"{'Resample':<{col_w}}" + f"{'Gaussian':<{col_w}}")
print("KS p>0.05:")
print(header)
print("-" * len(header))
gen_val = f"{(pvals_gen_oos > 0.05).mean():.1%}"
rs_val  = f"{(pvals_rs_oos  > 0.05).mean():.1%}"
gs_val  = f"{(pvals_gs_oos  > 0.05).mean():.1%}"
print(f"{'vs OOS':<16}{gen_val:<{col_w}}{rs_val:<{col_w}}{gs_val:<{col_w}}")
```

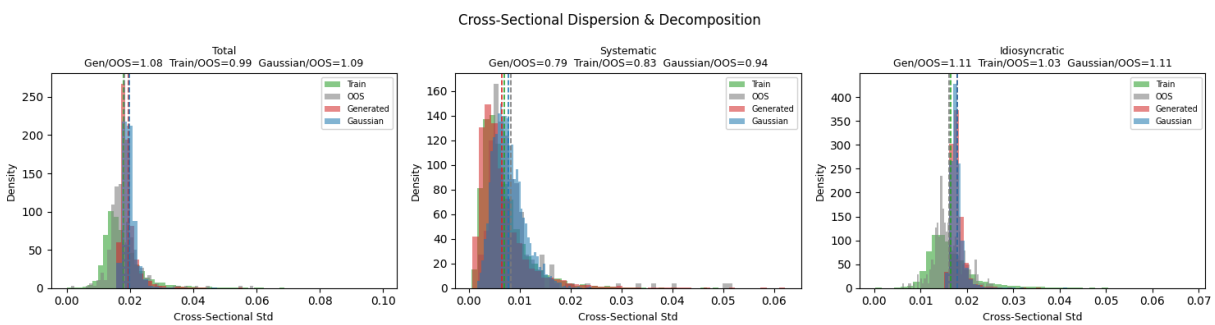
```
KS p>0.05:
                Generated           Resample            Gaussian
------------------------------------------------------------------------
vs OOS          56.5%               71.2%               42.9%
```

In [11]:
```
beta_c    = train_model.beta[:, col_indices] # beta
res_std_c = train_model.res_std[col_indices]
res_df_c  = np.asarray(train_model.res_df)[col_indices]

# training part construction
sys_train  = train_model.F.values.astype(np.float32) @ beta_c
idio_train = train_model.residuals[:, col_indices].astype(np.float32)
# test part construction
test_beta_c = test_model.beta[:, col_indices_test]
sys_oos  = test_model.F.values.astype(np.float32) @ test_beta_c
idio_oos = test_model.residuals[:, col_indices_test].astype(np.float32)
```

In [12]:
```
assert np.allclose(sys_train + idio_train, R_train[1:, :], equal_nan=True)
assert np.allclose(sys_oos + idio_oos, R_test[1:, :], equal_nan=True)
```

In [13]:
```
# helper: factor samples -> (sys, idio) for common stocks
def _decompose(sampler, num, model, beta_c, res_std_c, res_df_c, seed):
    fs_raw = sampler.generate(num).astype(np.float32)
    if model.factor_type == "regression":
        fs_full = np.column_stack([np.ones((len(fs_raw), 1)), fs_raw]).astyp
    else:
        fs_full = fs_raw
    sys = fs_full @ beta_c
    N, S = sys.shape
    rng = np.random.default_rng(seed)
    idio = np.empty((N, S), dtype=np.float32)
    for s in range(S):
        df_s = float(res_df_c[s])
        if np.isfinite(df_s):
            idio[:, s] = (rng.standard_t(df_s, size=N) * np.sqrt((df_s - 2)
        else:
            idio[:, s] = (rng.standard_normal(N) * res_std_c[s]).astype(np.f
    return sys, idio

sys_gen,      idio_gen      = _decompose(diffusionScenario.sampler, NUM_GENE
sys_gaussian, idio_gaussian = _decompose(gaussionScenario.sampler,  NUM_GENE

panels = [
    ("Total",         R_gen,    R_test,  R_train),
```

```
    ("Systematic",    sys_gen,  sys_oos, sys_train),
    ("Idiosyncratic", idio_gen, idio_oos, idio_train),
]

gs_mats = {
    "Total":         R_gaussian,
    "Systematic":    sys_gaussian,
    "Idiosyncratic": idio_gaussian,
}

fig, axes = plt.subplots(1, 3, figsize=(15, 4))
for ax, (comp, gmat, omat, tmat) in zip(axes, panels):
    gsmat = gs_mats[comp]
    gv  = np.nanstd(gmat,  axis=1)
    gsv = np.nanstd(gsmat, axis=1)
    ov  = np.nanstd(omat,  axis=1)
    tv  = np.nanstd(tmat,  axis=1)
    ax.hist(tv,  bins=50, density=True, alpha=0.55, color="#2ca02c", label="
    ax.hist(ov,  bins=50, density=True, alpha=0.55, color="#7f7f7f", label="
    ax.hist(gv,  bins=50, density=True, alpha=0.55, color="#d62728", label="
    ax.hist(gsv, bins=50, density=True, alpha=0.55, color="#1f78b4", label="
    for val, col in [(tv.mean(), "#2ca02c"), (ov.mean(), "#7f7f7f"),
                     (gv.mean(), "#d62728"), (gsv.mean(), "#1f78b4")]:
        ax.axvline(val, color=col, linestyle="--", lw=1.2)
    r_gen = gv.mean()  / (ov.mean() + 1e-12)
    r_tr  = tv.mean()  / (ov.mean() + 1e-12)
    r_gs  = gsv.mean() / (ov.mean() + 1e-12)
    ax.set_xlabel("Cross-Sectional Std", fontsize=9)
    ax.set_ylabel("Density", fontsize=9)
    ax.set_title(comp + "\nGen/OOS=%.2f  Train/OOS=%.2f  Gaussian/OOS=%.2f"
    ax.legend(fontsize=7)

fig.suptitle("Cross-Sectional Dispersion & Decomposition", fontsize=12)
fig.tight_layout()
plt.show()
```



In [14]:

```
row_idx_r = row_idx[row_idx < len(sys_train)]
sys_rs    = sys_train[row_idx_r]
idio_rs   = idio_train[row_idx_r]

rs_panels = [
    ("Total",         R_rs,    R_test),
    ("Systematic",    sys_rs,  sys_oos),
    ("Idiosyncratic", idio_rs, idio_oos),
]
```

```python
col_w = 20
header = (f"{'Component':<16}{'W1(Gen, OOS)':<{col_w}}"
          + f"{'W1(Gaussian, OOS)':<{col_w}}{'W1(Resample, OOS)':<{col_w}}")
print("Wasserstein-1 Distance of Cross-Sectional Std (vs OOS):")
print(header)
print("-" * len(header))
for (comp, gmat, omat, _), (_, rsmat, _) in zip(panels, rs_panels):
    gsmat = gs_mats[comp]
    gv  = np.nanstd(gmat,  axis=1); gv  = gv[np.isfinite(gv)]
    gsv = np.nanstd(gsmat, axis=1); gsv = gsv[np.isfinite(gsv)]
    ov  = np.nanstd(omat,  axis=1); ov  = ov[np.isfinite(ov)]
    rsv = np.nanstd(rsmat, axis=1); rsv = rsv[np.isfinite(rsv)]
    print(f"{comp:<16}{w1dist(gv, ov):<{col_w}.4f}{w1dist(gsv, ov):<{col_w}.
```

```
Wasserstein-1 Distance of Cross-Sectional Std (vs OOS):
Component       W1(Gen, OOS)        W1(Gaussian, OOS)   W1(Resample, OOS)
--------------------------------------------------------------------------
Total           0.0021              0.0029              0.0013
Systematic      0.0018              0.0010              0.0014
Idiosyncratic   0.0022              0.0022              0.0014
```